

Kavya Agar

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Summary: SWE with 2 internships and expertise in ML, data engineering, and full-stack web dev. Proven ability to deliver scalable software solutions, lead agile teams, and deploy production-ready systems using AWS, Docker, Kubernetes, and distributed ML.

EDUCATION

Texas A&M University, College Station, TX

Aug 2024 - May 2028

Bachelor of Science in Computer Science | GPA: 3.5

Minor in Statistics and Minor in Business | Dean's List

Relevant Coursework: DSA, Program Design, Discrete Mathematics, Computer Organization, Linear Algebra, Statistics

SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, TypeScript, Swift, SQL, R, Assembly

Frameworks & Libraries: React.js, Django, Flask, PyTorch, TensorFlow, Keras, HuggingFace, LangChain, Scikit-learn, Pandas, NumPy

Cloud & DevOps: AWS (EC2, S3, SageMaker, Step Functions), Kubernetes, Docker, Terraform, Git, GitHub Actions, OpenShift, CI/CD

Databases & Data Engineering: PostgreSQL, MongoDB, MySQL, Cassandra, DynamoDB, SQLite, ETL Pipelines, Data Warehousing

Machine Learning & AI: NLP, Deep Learning, MLOps, Model Deployment, ML Model Serving, Generative AI, LLM Fine-Tuning

Systems & Tools: Agile/Scrum, TDD, RESTful APIs, Microservices, Distributed Systems, Containerization, Postman, JIRA, UI/UX Design

EXPERIENCE

Aspen Technology, Houston, TX

Jan 2024 - Apr 2024

Business Analyst Intern

- Performed a **competitive analysis** on 5 **multivariate data analysis** tools to understand feature functions in relation to customer requirements
- Delivered actionable insights to **optimize** resource allocation across various industries through **market research and analysis**, with an estimated **15% increase** in total revenue
- Collaborated with cross-functional stakeholders (PMs, DSs) to evaluate collaboration workflows and improve product strategy for **olefin engineering software**

JPMorgan Chase & Co., Houston, TX

Jun 2023 - Aug 2023

Software Engineering Intern | Team Lead

- Engineered an **automated, scalable** data pipeline architecture in **Java**, migrating from **Apache Cassandra** to **AWS S3**, implementing custom Parquet conversion algorithms and **multithreading**, and improving data migration **efficiency by 90%**
- Led daily stand-ups for a team of 8, ensuring project milestones were met on time; presented the final solution to Executive Directors, earning **recognition** for its impact on data management

TECHNICAL PROJECTS

Financial NER Analyser | Python, PyTorch, HuggingFace, AWS (Step Functions, SageMaker, Textract), React, Docker

Sep 2025

- Fine-tuned DistilBERT on 116K+ tokens**, achieving **98% accuracy** and reducing manual document review time by **70%**
- Architected a **scalable AWS pipeline** using **Step Functions** to orchestrate **Textract** to **SageMaker** to a validation workflow, processing documents with automatic error handling and retry logic
- Built a production-ready document classifier** supporting multiple financial document types (bank statements, tax forms, loan applications) with custom entity recognition for amounts, dates, and account numbers

Stock Sentiment Dashboard | Python, NLP, AWS (Sagemaker, DynamoDB), HuggingFace, Terraform, Flask, React, Docker

Aug 2025

- Orchestrated an end-to-end machine learning pipeline using **Terraform** and **Docker** to automate model serving and data ingestion from financial news sources (e.g., Reddit, Finnhub)
- Developed a real-time NLP-based sentiment analysis system with **HuggingFace Transformers** and **VADER**, deploying the model on **AWS SageMaker** for scalable inference (**70% accuracy**)
- Designed a full-stack application with **Flask** and **React** to visualize sentiment, storing and serving live data with **DynamoDB**

FIN-ance | Python, TensorFlow, Scikit-Learn, React, OAuth, PostgreSQL, Django, SwiftUI, Recharts, Docker, Figma

Jun 2025

- Developed a **cross-platform expense tracking app** with **ML-driven spending forecasts**, improving budget prediction accuracy by **40% (90% model accuracy)**
- Implemented **OAuth 2.0 authentication** and a secure login system, enabling multi-user support and real-time syncing between iOS (**SwiftUI + WidgetKit**) and web clients
- Built a **Django REST API** for instant expense logging and syncing, backed by **PostgreSQL** and **Dockerized** for **scalability**

Comic Book Enhancer, TIDAL AI/ML Hackathon | Python, ML

Oct 2024

- Developed a **CNN-based image processing pipeline** to colorize black-and-white comic panels, achieving **85% accuracy** using deep learning; enhanced accessibility for color-blind users by optimizing color contrast and differentiation

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02 | Jun 2025)
- BCG GenAI Job Simulation - Forage (Aug 2025)
- Databricks GenAI Fundamentals (Aug 2025)
- IBM DevOps & Software Engineering - Coursera (Jul 2025)